Solar-Powered Additive Manufacturing in Extraterrestrial Environments

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ABSTRACT

Kickstarting a space economy will require building communication relays, refueling depots, repair depots, habitats, and mining bases from in-situ resources in strategic locations between Earth, Moon, and Mars. Due to the high costs inherent in transporting resources from the Earth's surface to these locations, new methods of material extraction and construction are necessary. Paramount to these development requirements is the need for a low-cost and efficient means for construction of habitats and physical structures. Utilizing networks of small spacecraft and robots to perform the task will reduce cost, enable scalability, and robustness. The idea of 3D printing structures has risen to the forefront of construction methods for its ability to be sent in advance of the primary mission and build structures autonomously. Two distinct challenges are inherent in this concept: the 3D printer needs to be supplied material, and it must have the ability to generate a significant amount of energy to process the material into its final form. Refining this printing technology to be as energy and resource efficient as possible is of the utmost importance to future space missions. Once this is achieved, it will be economical to build lunar and planetary bases rooting in in-situ resource utilization. Reducing the necessary supply of material to the additive manufacturing process and the power consumption leads to a reduction in the size of these early missions. In an effort to confront these challenges, we are working to develop an additive manufacturing process based on the principles of the selective laser sintering (SLS) technique, whereby a heat source (a laser in the case of SLS) heats the material just below its liquefaction point before returning to a solid form. By replacing the laser in the SLS process with a large Fresnel lens, we aim to focus enough sunlight to be able to sinter the material and create solid shapes. In this way, the system fully relies on renewable solar energy for its operation. In this paper, we propose development of solar additive manufacturing printers for melting and use of sand for construction. The paper will analyze the conceptualization, design, and prototype construction of the solar 3D printer. Current simulations are being done in support of the printer to determine best operating parameters and performance. Lesson learned from the simulations and prototype development will be used to develop a miniature scale printer for extended experiments.

INTRODUCTION

Space is now open for business. Accompanying this new market is a rising need for sustainable construction methods that can support a wide array of missions to extraterrestrial worlds. As both commercial enterprises and government space organizations plan long duration missions to deep space locations, a new economy is being developed that will require a vast spectrum of infrastructures. Communication relays, refueling depots, repair depots, human and

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robotic habitats, mining bases, and manufacturing facilities are just a few of the plethora of necessary structures. These infrastructures will need to be strategically located on the Moon, Mars, asteroids, other moons, and a host of other locations to support these deep space missions.

Due to the vast distance between these extraterrestrial locations where infrastructure is necessary, the time required to reach them, and the high export cost of materials from Earth, development of new methods of material extraction and utilization are paramount. These methods must be low-cost, efficient, and adaptable.

In recent years, additive manufacturing has quickly risen to the forefront of manufacturing techniques that are economically and technologically suitable for space infrastructure. Additive manufacturing techniques in use today are becoming mainstream in the arsenal of manufacturers, researchers, and developers. Recognizing their potential in space applications, NASA has hosted a 3D Printed Habitat Challenge for the past five years to allow organizations to compete for innovative additive manufacturing techniques for habitat construction. Companies such as Made In Space are deeply committed to the development of additive manufacturing techniques in the space environment.

Development of innovative additive manufacturing techniques designed to operation in extraterrestrial environments has vast potential for resource- and energy-efficient means of constructing the infrastructure necessary for space exploration. Current technologies need to be adapted to operate in these environments and new techniques should be explored to expand the capabilities and cost of deep space missions.

CHALLENGES TO EXTREME ENVIRONMENT ADDITIVE MANUFACTURING

Additive manufacturing in distant space environments is fraught with challenges. Among the most pressing of these concerns is the availability of the materials needed for the manufacturing process. The feasibility of importing the necessary materials for large-scale infrastructure comes at such a high cost, it will be disregarded for the purposes of this discussion. That leaves in-situ resources as the one, feasible option to cut costs. However, the material will still need to be extracted, separated into useable constituents, and refined. The exact refinement methods needed depends on the printing process being employed, but would typically involve pelletizing it to a useful product in Fused Deposition Modeling (FDM) style printing operations.

Availability of material suitable for the printing process is one of the prime concerns of a printing operation on other worlds. With the high cost of importing materials, in-situ resource utilization (ISRU) is of the utmost importance to avoid high fuel expenditures, logistics, and time delays. To be successful at ISRU, the printing operation must also be able to refine the material into a printable form.

Printing techniques that can only work with a narrow band of materials in turn narrow their own scope of usefulness and the environments where they can be employed. Thus, a printing process that can work with a variety of materials or a composite material is optimal.

Power systems for any spaceborne vehicle are another critical aspect of the operation. This is especially true for an additive manufacturing operation in the deep-space environment. In addition to the power requirements of the electrical systems, communication, heating, and other systems, power must be allocated to the operation of the printing operation. Gathering material, refinement of material, and heating the processed material are all energy-intensive operations that must be accounted for. Designing sources of power for these operations can be a significant challenge in these remote locations.

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OBJECTIVE

In order to combat these challenges, we are working to develop a method of additive manufacturing tailored to off-world environments. Our prime constraints in the development of this technology are that it must:

1. Be low-cost and low-energy.
2. Be adaptable to a variety of extraterrestrial environments.
3. Adhere to the concept of In-Situ Resource Utilization (ISRU).
4. Be suitable for an autonomous printing operation.

We believe that it is possible to develop or adapt an additive manufacturing process sufficiently for use in extreme extraterrestrial environments like the Moon, Mars, and asteroids. By using the framework and fundamental concepts of current additive manufacturing techniques, we can adapt certain parameters to further their functionality in unique, new environments.

OTHER APPROACHES

A multitude of companies, organizations, universities, and research labs have worked to develop concepts for additive manufacturing in extraterrestrial environments. A number of startups have gained a spotlight on their work by competing in NASA's Mars Habitat Challenge. Team Zopherus from Arkansas, won Phase 3: Level 1 in 2018 (Rowe 2020). Their concept featured a mobile 3D printing platform that uses deployable rovers to gather resources. The AI SpaceFactory design agency won first place in the finale of the challenge with their MARSHA habitat (Rowe 2020) that uses a large 3D print head to construct a uniquely-shaped structure out of a basalt fiber material (Figure 1). Team SEArch+/Apis Cor won first place in the Phase 3: Level 4 on-site construction portion of the challenge employing a radial FDM print mechanism that they also use to 3D print terrestrial homes (Rowe 2020).

Figure 1. AI Spacefactory employing their basalt fiber FDM print head during the NASA 3D Printed Habitat Challenge (Malott, 2019)

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Other efforts outside of the NASA challenge have made strides in developing additive manufacturing in extreme environments in innovative ways. Made In Space developed a 3D printer with minor modifications to be used aboard the International Space Station (Snyder et al. 2013).

Alexandre Meurisse of the European Space Agency demonstrated the ability to sinter artificial lunar regolith using an array of focused sunlight and Xenon lamps (Meurisse 2018). The use of artificial light sources allowed for more consistent beam of light such that they were able to create a brick layer-by-layer from the regolith. He was able to create a three-dimensional object out of sintered material.

In 2011, Markus Kayser of the Royal College of Art in London utilized the power of the sun to sinter sand in the Moroccan desert and make intricate shapes (Figure 2) (Kayser 2011). He replicated the basis of the Selective Laser Sintering (SLS) process by focusing the desert sunlight with a large Fresnel lens. Using this focused energy, he was able to bring the sand to its sinter temperature and create a glassified structure layer-by-layer. His artistic works provided a clue into how we might be able to overcome many of the challenges associated with extraterrestrial additive manufacturing.

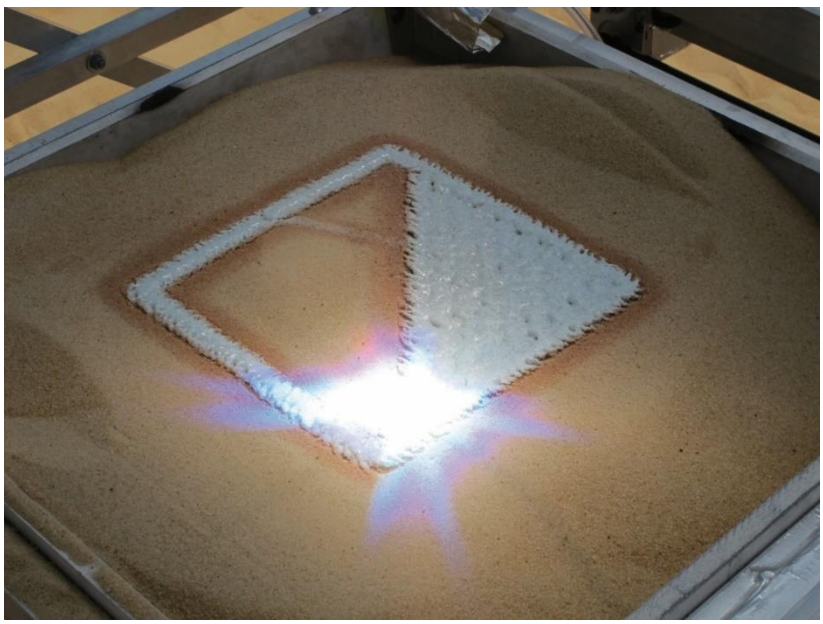

Figure 2. Markus Kayser's solar sinter project he put to work in the Moroccan desert.

DESIGN METHOD

The majority of additive manufacturing (AM) techniques that have been explored for use on other worlds have been centered on the Fused Deposition Modeling (FDM) approach. The FDM process has the capability, as shown by many of the NASA Habitat Challenge to meet our objectives of adherence to ISRU and the ability for autonomy. However, this technology falls short in comparison to other methods in terms of energy usage and adaptability.

Selective Laser Sintering (SLS) is promising new approach to extraterrestrial additive manufacturing that has potential to meet all of our stated design objectives. Because the SLS printer can be positioned on the surface where it is needed, it is an adaptable technique. In many

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cases, it requires minimal pre-processing of materials. In Markus Kayser's sinter project, he printed structures directly into the sand of the desert. By varying the output of the focused energy source, it is adaptable to a variety of print materials. All that is needed is knowledge of the sinter temperature of the print material and the energy source heats the material.

The energy requirements of the SLS process is attractive as well when compared to others. A study at the University of Illinois at Chicago (Yang et al. 2017) compared specific energy consumption among multiple additive manufacturing processes and found that of SLS to be far lower than FDM when printing with polymeric materials. In the study, SLS had a specific energy consumption near 40 kWh/kg while FDM consumed 115 kWh/kg. However, material selection is a significant factor in the comparison of printed processes. The standard FDM process requires the material to reach its melting point before being extruded through a nozzle. The SLS process only requires the material to reach a sinter temperature, which, in some cases, may be drastically lower. To understand the differences, it is important to know a bit about the SLS process.

Typical SLS technology makes use of a laser guided by a mirror that heats a powder substrate to its sinter temperature, creating a fused structure layer-by-layer (Figure 3). The sinter temperature for most materials is at least a few hundred degree Celsius below the actual melting point, so no state change is necessary for sintering. An SLS printer first raises the chamber temperature to near the sinter temperature of the material. Next, the laser is guided over the leveled powder material in the print bed. Once that layer is complete, the print bed is lowered, and thin powder layer is added, and the process repeats. In this way, the print bed provides the support structure to the part and has the ability to create intricate shapes (included embedded parts).

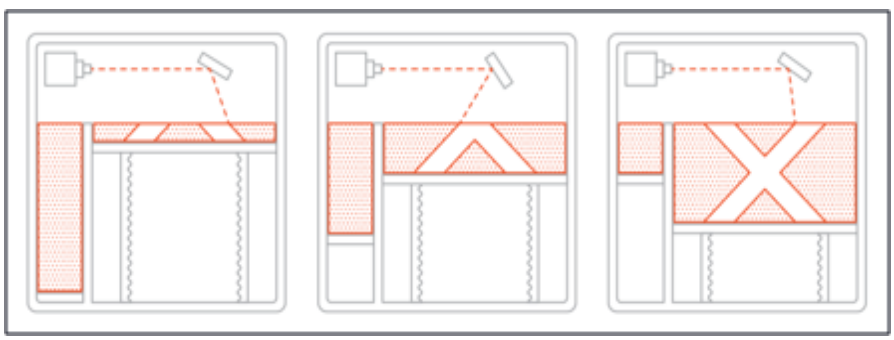

Figure 3. Typical SLS process used a laser guided over the surface of the print bed. Print bed is advanced (negative Z direction) as a new layer is leveled on top (Varotsis 2020).

Fuel and energy sources in some remote environments are scarce and can require complex refinement processes. In order to achieve our prime objective of reducing the energy needs of a printing operation, we need to further reduce the energy demand of the SLS process. We need a printing method that harnesses available resources to operate. By replacing the CO2 laser onboard an SLS printer with a properly sized Fresnel lens, we can all but eliminate the energy requirements of the printing process by utilizing a renewable energy source. In areas where sunlight is available in sufficient quantities, so is power source that can be harnessed to produce the necessary increase in thermal energy to sinter in-situ materials.

A Fresnel lens is designed to remove excess weight and material from a standard lens, while preserving the optical qualities and performance (Figure 4). With the unnecessary interior

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material removed from a convex lens, the optical surface, which provides the lens with is characteristic properties is planted on a flat surface. This leaves us with a far more economical lens for use in the solar sinter system.

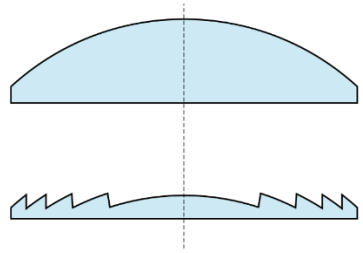

Figure 4. A Fresnel lens reduces weight by removing interior material while preserving the optical surface.

DESIGN CONSIDERATIONS AND CHALLENGES

While it shows promise to reducing energy usage, there are significant considerations and challenges to take into account when replacing a laser with focused sunlight in the SLS process. As we confirmed through basic feasibility testing, there are inconsistencies associated with focused solar energy compared to a laser. In environments that have an atmosphere, it will always be a source of variable conditions that need to be contended with to be successful in harnessing as much solar energy as possible. As photons interact with gas molecules in an atmosphere, they are absorbed and scattered. This can impact a solar sinter printer on the surface that is harnessing energy from the sunlight. A passing cloud can block the sunlight reaching the lens and disturb the power output of the focused beam, leaving pockets of unsintered material. A solar SLS printer would either need the ability to adapt to these variances to maintain as consistent of a beam as possible or limit its use to suitable weather conditions.

Successful employment of solar additive manufacturing techniques is highly dependent on how much sun the machine will receive. This will effectively rule out printing operations in locations where the body is not at least partially exposed to the sun. Printing operations are restricted to daylight hours unless a backup source of energy is available.

In contrast, a CO2 laser provides a consistent beam during any light conditions with variable power and beam width. With a focused solar beam, the width is a function of the ratio of apparent height and the height of the lens above the print surface—assuming an infinite distance to the Sun, as is the case for most calculations of this sort. This means that the focal area is maintained only by keeping the lens at a consistent height.

Additional factors provide challenges to a solar sinter in extraterrestrial environments. For one, the absence of a pressurized system. Sintering requires heat and pressure, so printing in a vacuum or near vacuum may provide challenges to this process. Also, the lack of ambient temperature control can pose challenges. A traditional SLS printing machine will bring the ambient temperature of an enclosed chamber to just below the sinter temperature. This allows the use of a lower power laser to heat the powder substrate the rest of the way to the sinter temperature.

Despite these challenges, we know that the concept has been proven before using focused sunlight on Earth. In at least one case even reaching the necessary temperature to sinter Moroccan desert sand. This provides evidence that the concept is sound and can be controlled in

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a way to provide the means for a low energy, adaptable, and potentially autonomous printing operation using in-situ resources. This is the technology we are working to develop.

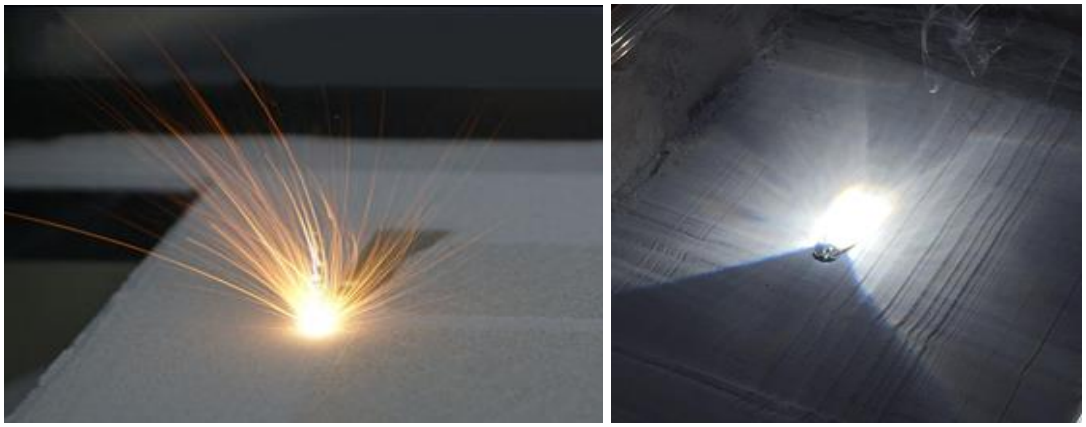

Figure 5. (left) Selective Laser Sinter process uses a CO2 laser. (right) SpaceTREx team demonstrated focused solar beam to sinter powdered

FEASIBILITY AND CALCULATIONS

To determine the feasibility of the solar sinter process in extraterrestrial environments, we must first investigate the amount of available solar power at various locations in the solar system. Based on the average closest approach, the average solar irradiance is shown in Figure 6. Given that the Mars system has approximately half the solar irradiance available in the Earth/Moon system, certain steps may be necessary to bolster the energy available for printing. Some of these steps will be considered later in this paper.

	Sun Approach Distance (AU)			Solar Irradiance (W/m^2)		
Object	Closest	Farthest	Mean	Max	Min	Mean
Mercury	0.307	0.466	0.387	14656	6361	9247
Venus	0.718	0.728	0.723	2679	2606	2643
Earth/Moon	0.980	1.010	0.995	1438	1354	1395
Mars	1.380	1.660	1.520	725.3	501.3	597.9
Asteroid Belt	2.200	3.200	2.700	285.4	134.9	189.5
Jupiter	4.950	5.460	5.205	56.38	46.34	50.99
Saturn	9.050	10.120	9.585	16.87	13.49	15.04
Uranus	18.400	20.100	19.250	4.080	3.419	3.728
Neptune	29.800	30.400	30.100	1.555	1.495	1.525
Pluto	29.700	49.300	39.500	1.566	0.568	0.885

Figure 6. Solar irradiance available at various locations throughout the solar system.

Making use of the Lens equation (Equation 1) and the Magnification equation (Equation 2), we can calculate the lens power available at these locations.

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$$\frac{1}{f} = \frac{1}{d_o} + \frac{1}{d_i} \tag{1}$$

$$M = \frac{h_i}{h_o} = -\frac{d_i}{d_o} \tag{2}$$

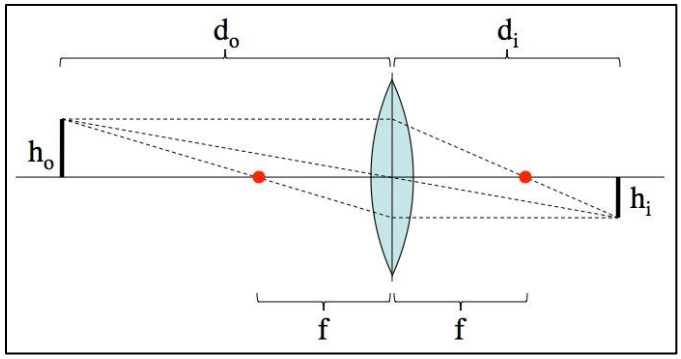

Figure 7. Visualization of the lens calculations and equations.

Based on observational data, we know the apparent height of an object (the Sun in this case) when standing on the surface of any celestial body. When this value is compared to the mean distance of the object from the Sun, we have the following ratio in the Earth/Moon system:

$$\frac{h_o}{{}^{o}d} = \frac{1}{110} \; where\ h_o is\ the\ height\ of\ the\ object\ and\ d_o\ is\ the\ distance\ to\ the\ object.$$

This ratio can be calculated for any location where the available power is being calculated. We can utilize this ratio with the Magnification equation to determine the height of the projected image (hi). This calculation assumes that the Sun is at an infinite distance from the lens (do), therefore it can be assumed that the distance to the projection is equal to the focal length (d_i = f).

$$\frac{h_o}{d_o} = \frac{1}{110} \; (on\ Earth)$$

$$\frac{h_i}{h_o} = \frac{d_i}{d_o} = \frac{1}{110}$$

$$d_i = f \; (assuming\ Sun\ is\ at\ infinate\ distance)$$

$$h_i = f \cdot \frac{1}{110} \; (height\ of\ focal\ point\ on\ Earth)$$

$$A_f = \pi \cdot (\frac{h_i}{2})^2 \; (area\ of\ focal\ point)$$

$$P_{lens} = Irradiance\ [W/m^2] \cdot Lens\ Efficiency \cdot Lens\ Collection\ Area$$

We can also calculate the achievable power of the lens using the irradiance from Figure 8, the lens efficiency, and the collection area of the lens (easy to calculate with a rectangular Fresnel lens). With a conservative estimate of 50% lens efficiency and a 1 m^2 lens, we can

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calculate the power we could focus at these locations. Three locations were chosen here due to their higher solar irradiance and necessity for space infrastructure. It should be noted that the calculations for achievable power on the Earth's surface use an irradiance of 1000 W/m^2 (due to atmospheric effects), which is why the total power is less than on the Moon.

Location	Achievable Lens Power (W)
Earth Surface	393
Moon	548
Mars	235

Figure 8. Achievable lens power (W) at several locations. Calculations at Earth's surface assumes a solar irradiance of 1000 W/m^2.

Knowing the lens power we could produce for sintering at these locations, we now must compare to the estimated power requirements to sinter available materials. For purposes of this conceptual development, we chose to follow a similar series of calculations (Lobanov 2013) used by the additive manufacturing industry to calculate the power requirements of SLS printing lasers. The process requires the input of material properties, ambient conditions, beam size, travel speed, and other parameters to calculate the power requirements to sinter the material (Figure 9).

Location	Material	Required Sinter Power (W)	Feed Rate (mm/s)
Earth Surface	Sand	351	2.50
Moon	Regolith	450	1.00
Mars	Basalt	236	4.50
Deimos	Carbonaceous Chondrite	97	5.00

Figure 9. Power requirements and feed rates for sintering in-situ materials.

The resulting calculations show that dependent on the printing parameters used, there is ample power to sinter in-situ materials. The resulting power requirements are highly dependent on the feed rate—if the printer can slow down the feed/travel rate enough, it will allow more time for the material to heat and reach its sinter temperature. Other aspects of the physical environment like the rotation of a smaller body may make long exposures difficult to achieve, the suitable locations will provide the necessary power. By tuning a multitude of factors in both design and operation, a printing operation could be possible. Adapting lens size, locating printing operation in areas with high solar exposure, slowing feed/travel rates, and scheduling print operations are all ways that the operation can be tailored to its environment and produce the necessary power to sinter.

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EMPLOYMENT OF TECHNOLOGY AND TECHNICAL CONSIDERATIONS

As discovered in the preceding sections, there are some limitations to the employment of the solar sintering process for additive manufacturing. In certain locations where the right materials are available or solar irradiance is not at a premium, it may be more effective and economical to employ more traditional FDM printing processes. In the same regard, there are certainly locations where the converse is true. In locations where power source options are limited, material refinement is impractical, or cost is a factor, solar sintering may be able to fill a niche role in the development of space infrastructure.

Because of the slow feed rates necessary, solar sinter technology may be best employed as part of a robotic fleet tasked with autonomous construction prior to human habitation. In order for an autonomous solar sinter operation to feasibly construct the infrastructure necessary for bases, the technology should be employed on a mobile platform. With the Fresnel lens sinter process loaded on the back of a rover, it would sinter the in-situ materials directly onto the surface of the planet, moon, or asteroid as it traversed the terrain. A blade fixed to the front of the rover, very similar to a road grader, would level the surface as the rover traveled, providing a clean surface for the sinter process. The combined level and sinter process by a robotic fleet is visualized in Figure 10.

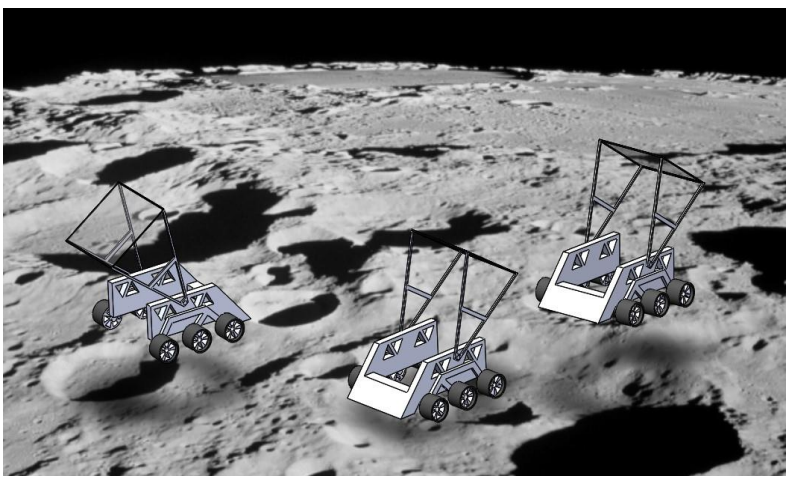

Figure 10. Visualization of an autonomous fleet of sinter robots on an extraterrestrial body.

Certain infrastructures will be better suited applications of the solar sinter technology due to the printing process. An SLS printing operation involves z-axis translation of the print bed as new layers are formed on top of the previous. If the solar sinter is applied in a similar manner, it would have similar capabilities given a print bed and a source of material. However, for the solar sinter process to be applied to the construction of larger infrastructures, the printing operation must be conducted directly onto the surface where the material is available. New layers of powdered regolith would need to be formed and leveled on top of the previous layer.

When printing directly on to the surface of a planet or asteroid, certain categories of infrastructures might be a more gainful employment of the technology. Building foundations, road networks, and structural pieces that can be assembled later to form vertical structures are several applications where the solar sinter process might be more effective. Taller structures require a vertical travel that might be difficult to build in layers.

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EXPERIMENTAL OBJECTIVE

To prove the feasibility of the concept, we are working to develop a 3D solar sinter machine capable of sintering typical quartzite sand. In order to be successful, the experimental setup must reach approximately 1100˚C using a Fresnel lens. The solar sinter will have the capability to accept a variety of different materials for testing purposes. Our objective is to use the machine to verify the efficacy of the calculations and concept in as many simulated environments as possible. By subjecting the printing process to a host of ambient temperatures, lighting conditions, and even variable pressures using our vacuum chamber, we can help to gain a better understanding for the potential uses of the technology.

We conducted an early feasibility testing using a small 8.5” x 11” Fresnel lens and sinter powder with a low melting point (Figure 11). With this basic setup, we were easily able to reach the required sinter temperature of the material and create rough 3D structures by manually adding new layers.

Figure 11. Early feasibility testing verified sinter process on a powdered substrate. Multiple layers were joined to make a 3D structure.

EXPERIMENTAL DESIGN

To begin testing the concept, we are working to complete a mid-scale design (75cm by 75cm) before completing the full-scale design. Being the approximate size of a large 3D printer, the mid-scale design will grant us greater flexibility to conduct environmental testing since it is sized appropriately to fit in our lab’s vacuum chamber. Here, it can be subjected to lower pressures and chilled to low temperatures. This proof-of-concept design will also allow us the ability to test functional aspects of the printer concept, such as print bed control, sun tracking, and motor sizing. The full-scale design will use a 1 m^2 Fresnel lens.

In a typical SLS printer, the print bed only travels in the Z-axis, while the laser is guided by a mirror so X- and Y-motion are not necessary. For this concept, the machine will work to keep the beam focused on a fixed point and the print bed will move in the X and Y directions based on Gcode commands to make a design. As each layer is completed, a new layer of powdered substrate will be leveled on top and the print bed will move in the negative Z-axis before starting the new layer.

Travel for the print bed in the X- and Y-axes will be by a pair of motors assigned to each axis. These motors will drive a belt that slides a carriage along their respective axes. Z-axis motion will be accomplished by two verticle screws powered by another pair of motors.

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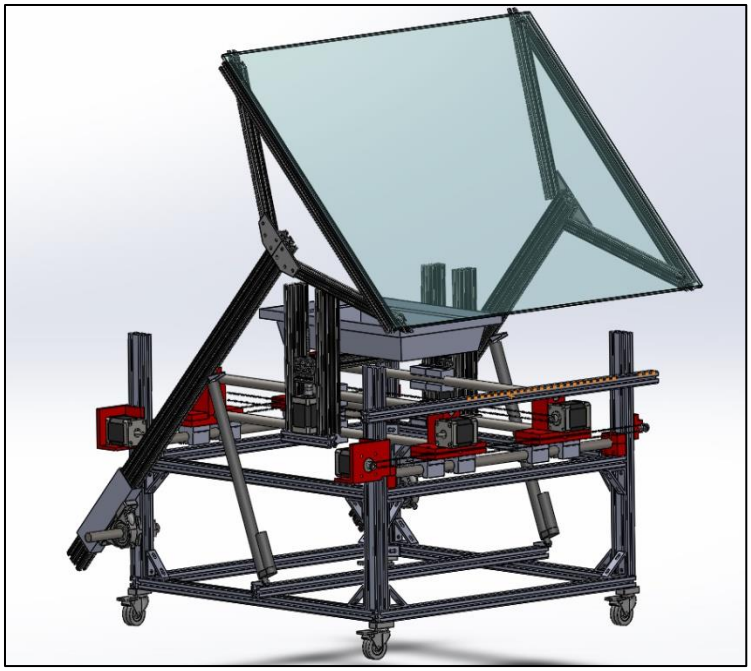

Figure 12. Solar sinter experimental design.

The heart of the solar sinter design will be the Fresnel lens providing the source of power to sinter the powdered substrate. In order to accomplish sinter temperatures, the lens will need to be maintained in an optimal position based on the location of the sun. The lens will be on a retracting arm so it can be focused. Based on the input from a simple Arduino-controlled sun tracker, the lens also has the ability to be tilted via a pair of linear actuators visible in Figure 12. The machine can also be mounted on a rotating, controlled platform that allows the machine to rotate and provide more efficient sun tracking.

With the advent of open source 3D printing, creating the electrical schematics and writing a control system for this machine is relatively simple. With an Arduino Mega and a Ramps 1.4 board providing operational control of the motor actions, an open source platform will allow us to feed the printer Gcode that is generated from a slicer program in the same way any other 3D printer would. Aside from the sun tracking for the lens, the control for the print bed is almost identical to other commercial or DIY 3D printers.

EXPERIMENTAL CHALLENGES

The experimental setup discussed above will have a number of challenges at each scale of the project. Each scaled solar sinter machine will have a set of strengths and weaknesses. Foremost to these challenges will be the ability to simulate extreme extraterrestrial environments large enough to fit the full-scale design. In an environmental test chamber and additional constraint is the lack of sunlight, so an artificial sun source will need to be included.

One of the key outcomes of testing will be a better understanding of how to scale a printer and still achieve optimal performance. Creative solutions including reflective solar collectors, more efficient lenses, and an advance solar tracker will be necessary to keep the technology viable in as many environments as possible.

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CONCLUSION

As we expand our human and robotic presence in the far reaches of the solar system, there is an accompanying demand for space infrastructure. Communication relays, mining bases, habitats, and a multitude of other structures are required to support bases and colonies. To make these infrastructures a reality, improved methods of construction that are low-energy and low-cost. Additive manufacturing is one of the most promising methods for construction in the extreme environments where this infrastructure will be located. It offers an adaptable platform where the structures that you need can be constructed without complex assembly and even human intervention. By adapting additive manufacturing techniques to use in-situ materials, they can greatly reduce the energy and cost input to the construction. While a myriad of technologies have been proposed for extraterrestrial construction, most have been concentrated on the Fused Deposition Modeling (FDM) process which has a number of inefficiencies in itself—namely the energy intensive pre-processing requirements. Other methods need to be developed that are tailored to work in applications where power conservation is a priority and material processing is not an option.

Solar sintering has shown that it can be a more practical option in certain locations and environments. With refinements in the process and machinery, it has promise to further reduce the energy demand of an autonomous 3D printing operation and potentially operate in environments that might not be accessible or cost-effective for FDM printing. By adjusting print parameters like beam travel rates, the lens aperture, and focal points, a solar sinter offers opportunities to tune the sinter power depending on the material and environment where it is operating. High versatility and low operating energy are among the key features that make the sinter technology worth further exploration and development.

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